



Pietro Ferrara

ASSOCIATE PROFESSOR

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Career

Ca' Foscari University

Venice, Italy

ASSOCIATE PROFESSOR

From Nov. 20219 (Tenure-track assistant professor until Nov. 2022)

- Research: I am actively involved in the development of the research activity of the Software and System Verification group (<https://ssv.dais.unive.it/>) writing and contributing to scientific papers (right now around 5 conference and journal articles are under evaluation), research proposals (for different funding schemes such as Horizon 2020, and national projects), and supervising postdocs and PhD students supervisions.
- Teaching: I am currently teaching three courses, Software Architecture (1st year of the master programme in Computer Science, about 40 students), Object oriented programming (2nd year of the bachelor programme in Computer Science, about 150 students), and Introduction to Coding and Data Management (1st year of the bachelor course in Digital Management, about 100 students). In addition, I am part of the teaching committee of the bachelor programme in Digital Management, and of the PhD programme in Computer Science.
- Entrepreneurship and industrial collaborations: I am developing together with some colleagues at Ca' Foscari University Secura Factors, a spin-off focused on the application of formal methods to robotic and industrial IoT software.

JuliaSoft

Verona, Italy

HEAD OF RESEARCH AND DEVELOPMENT AND RESEARCH SCIENTIST

Feb. 2016 - Nov. 2019

- My main goal was investigate and extend the barriers of science in order to deliver innovative solutions to deliver higher quality tools than our competitors.
- Extend and improve Julia's analyses to new programming languages (e.g., .NET [FCS18]), novel properties (e.g., leakages through intents [SFS18] and privacy analyses [FOS18]), and making the overall framework more flexible and expressive (e.g., through framework specifications [NF18]). This ranged from the design and mathematical formalization of the new features, their implementation and commercialization, and the academic dissemination of these results through scientific articles and talks.
- Create and maintain a network of scientific collaborations, and design and develop scientific projects from the initial proposal to their development.
- Supervision of interns for their stage and bachelor or master theses.
- Participate to the scientific community by being a program committee member of various international conferences and workshops, reviewing papers for several conferences and journals, participating and giving talks in international conferences, and giving lectures at different universities both for bachelor and master courses, and for scientific seminars.

IBM Thomas J. Watson Research Center

Yorktown Heights, NY, U.S.A.

RESEARCH STAFF MEMBER

Jul. 2013 - Dec. 2015

- Research: Apply static and dynamic analysis to security properties of mobile software (mostly privacy properties in Android) by developing various tools to detect violations at compile or run time. This effort already lead to about 10 publications (one awarded as an ACM Distinguished Paper), and an IBM Research Accomplishment Award.
- Community: Program committee member of 2 conferences, demo co-chair at OOPSLA 2015, reviewer for 10 international conferences or journals, interviewer of various Software Engineer and Research Staff Member candidates at IBM Research.

ETH Zürich

Switzerland

LECTURER AND POSTDOC

Apr. 2009 - Jul. 2013

Supervisor: Peter Müller (Full professor, ETH Zürich)

- Research: Design and implement a novel generic static analyzer (Sample) combining various value and heap analyses. 4 PhD and 6 master/bachelor students extended Sample with novel analyses. Overall, this effort lead to 10 publications in international journals and conferences. Sample currently supports various languages (namely, Scala, Chalice, and TouchDevelop), and many value (e.g., numerical) and heap (e.g., pointer and shape) analyses.
- Funding: Actively participate to the writing of two project proposals (600'000CHF in total, equivalent to about 580'000\$), and to their development as technical head and co-principal investigator.
- Teaching: Lecturer of a master course on static program analysis. Teaching assistant (teaching exercise sessions and evaluating final written exams or projects) various times for 3 different courses (both at bachelor and master level) on object-oriented programming languages, formal methods, and program verification. Supervise 3 master and 2 bachelor theses.
- Community: Program committee member of 3 conferences, external reviewer for 20 international journals and conferences, interviewer of about 30 candidates for PhD positions at ETH.

Education

Ecole Polytechnique, and Ca' Foscari University

Paris, France and Venice, Italy

PHD DEGREE IN COMPUTER SCIENCE

Oct. 2005 - May 2009

Subject: "Static analysis via abstract interpretation of multithread programs".

Advisors: Radhia Cousot (research director, CNRS/Ecole Normale Supérieure) and Agostino Cortesi (full professor and head of the Department of Computer Science, Università Ca' Foscari of Venice).

Reviewers: Manuel Hermenegildo (full professor, Universidad Politécnica Madrid) and Helmut Seidl (full professor, Technische Universität München).

Examiners: Eric Goubault (research director, CEA) and Francesco Logozzo (researcher, Microsoft Research).

Defended at Ecole Normale Supérieure of Paris on May 22nd, 2009.

Ca' Foscari University

Venice, Italy

MASTER DEGREE IN COMPUTER SCIENCE

Sept. 2003 - Feb. 2005

Grade: 110/110 cum laude.

Subject: "Development of a static analyzer for object oriented languages and application to the firewall analysis of JavaCard".

Advisor: Prof. Agostino Cortesi, Co-advisor: Dr. Francesco Logozzo (Ecole Polytechnique, Paris).

Ca' Foscari University

Venice, Italy

BACHELOR DEGREE IN COMPUTER SCIENCE

Sept. 2000 - Jul. 2003

Grade: 110/110 cum laude.

Liceo Scientifico Statale "G. Bruno"

Venice, Italy

HIGH SCHOOL DEGREE - STUDIES OF SCIENCE

Sept. 1995 - Jul. 2000

Grade: 94/100.

Research Interests and Vision

My research interests are focused on the application of rigorous mathematical theories to enhance the reliability, security, and performances of software by means of static analysis. Abstract interpretation is a framework applied to develop sound static analyses proving properties on all possible executions of a program. However, approximation is necessary to achieve computability, and the analysis might produce false alarms. Finding a good balance between precision, efficiency, and soundness depends on specific applications, and it usually requires deep research investigation. I am particularly interested in new scenarios and practical applications where static analysis might have a relevant impact. Right now, I am particularly keen to apply these theories to modern software architectures (e.g., microservices and serverless applications).

Keywords: abstract interpretation, static analysis, software engineering, program verification, microservices, security.

Entrepreneurial Activity

Winner in the ICT category with the project FACTORS: Formal Assurance of Cybersecurity and

Oct. 2020 **Trustability Of Robotics Systems (with Gianluca Caiazza, Ruffin White, Agostino Cortesi, and Sanna Kallio)**, Start Cup Veneto

Padova, Italy

Awards, Honors and Fellowships

March 2025 **Habilitation for full professorship (ING-INF 05 scientific area)**, ASN, Italian Ministry of University and Research

Italy

Feb.-Mar. 2025 **Invited Professor**, Sorbonne Université, Computer Science Laboratory

Paris, France

Oct. 2024 **Finalist for the Best Safety, Security, and Rescue Robotics paper award**, IROS

Abu Dhabi, United

Arab Emirates

Gothenburg,

Sweden

Jun. 2018 **ACM SIGSOFT FME Distinguished Paper Award**, FormaliSE

Oct. 2015 **IBM Research Accomplishment**, "Fundamental Contributions to Program Analysis and Programming Languages for Mobile"

U.S.A

Jul. 2014 **ACM SIGSOFT Distinguished Paper Award**, ISSTA

San Jose, U.S.A

Jul. 2011 **Invited**, Microsoft Research Faculty Summit

Redmond, U.S.A

Oct. 2008 **3 months Fellowship**, French Ministry of Foreign Affairs

France

May 2006 **Vinci Fellowship**, Italian-French University "italo-francese"

Feb. 2006 **2nd place**, Clusit Security Summit

Milan, Italy

JOURNALS

- [ZCF24]** G. Zanatta, G. Caiazza, P. Ferrara, L. Negrini: *"Inference of Access Policies through Static Analysis"*, International Journal on Software Tools for Technology Transfer, 2024
- [OAC24]** L. Olivieri, V. Arceri, B. Chachar, L. Negrini, F. Tagliaferro, F. Spoto, P. Ferrara, A. Cortesi: *"General-Purpose Languages for Blockchain Smart Contracts Development: A Comprehensive Study"*, IEEE Access, 2024
- [FAC24]** P. Ferrara, V. Arceri, A. Cortesi: *"Challenges of software verification: the past, the present, the future"*, International Journal on Software Tools for Technology Transfer, 236(4): 421-430, 2024
- [ONA24]** L. Olivieri, L. Negrini, V. Arceri, B. Chachar, P. Ferrara, A. Cortesi: *"Detection of Phantom Reads in Hyperledger Fabric"*, IEEE Access, 2024
- [NAC24]** L. Negrini, V. Arceri, A. Cortesi, P. Ferrara: *"Tarsis: An effective automata-based abstract domain for string analysis"*, Journal of Software: Evolution and Process, 2024
- [FH24]** P. Ferrara, L. Hadarean: *"State of the art in program analysis"*, International Journal on Software Tools for Technology Transfer, 26(5): 569-571, 2024
- [FAC24]** V. Arceri, L. Negrini, L. Olivieri, P. Ferrara: *"Challenges of software verification"*, International Journal on Software Tools for Technology Transfer, 26: 669-672, 2024
- [CFL23]** S. Calzavara, P. Ferrara, C. Lucchese: *"Certifying Machine Learning Models Against Evasion Attacks by Program Analysis"*, Journal of Computer Security, vol. 31, no. 1, pp. 57-84, 2023
- [FOS21]** P. Ferrara, L. Olivieri, F. Spoto: *"Static Privacy Analysis by Flow Reconstruction of Tainted Data"*, Int. J. Softw. Eng. Knowl. Eng. 31(7): 973-1016 (2021)
- [FMC21]** P. Ferrara, A. Mandal, A. Cortesi, F. Spoto: *"Static Analysis for Discovering IoT Vulnerabilities"*, International Journal on Software Tools for Technology Transfer, 23(1): 71-88, 2021
- [FCS20]** P. Ferrara, A. Cortesi, F. Spoto: *"From CIL to Java-bytecode: Semantics-based Translation for Static Analysis Leveraging"*, Science of Computer Programming, Volume 191, June 2020 (ERA: A)
- [FMCb19]** P. Ferrara, A. Mandal, A. Cortesi, F. Spoto: *"Cross-Programming Language Taint Analysis for the IoT Ecosystem"*, Electronic Communication of the European Association of Software Science and Technology, Volume 77, October 2019
- [SBE19]** F. Spoto, E. Burato, M. D. Ernst, P. Ferrara, A. Lovato, D. Macedonio, C. Spiridon: *"Static Identification of Injection Attacks in Java"*, ACM Transactions on Programming Languages and Systems, Volume 41 Issue 3, July 2019 (ERA: A*, MS PL: 2nd/268)
- [MPC19]** A. Mandal, F. Panarotto, A. Cortesi, P. Ferrara, F. Spoto: *"Static Analysis of Android Auto Infotainment and ODB-II Apps"*, Software: Practice and Experience, Volume 49, Issue 7, Pages 1131-1161, July 2019 (ERA: A, AM Journal: 56th/376)
- [CFH18]** A. Cortesi, P. Ferrara, R. Halder, and M. Zanioli: *"Combining Symbolic and Numerical Domains for Information Leakage Analysis"*, Transactions on Computational Science, Volume 31, pages 98-135, 2018
- [Ferr16]** P. Ferrara: *"A generic framework for heap and value analyses of object-oriented programming languages"*, Theoretical Computer Science, Volume 631, pages 43-72, June 2016 (ERA: A)
- [CFC15a]** A. Cortesi, G. Costantini, and P. Ferrara: *"The Abstract Domain of Trapezoid Step Functions"*, Computer Languages, Systems & Structures, Volume 43, pages 41-68, October 2015
- [CFC15]** G. Costantini, P. Ferrara and A. Cortesi: *"A Suite of Abstract Domains for Static Analysis of String Values"*, Software: Practice and Experience, Volume 45, Issue 1, pages 245-287, February 2015 (ERA: A, AM Journal: 56th/376)
- [Ferr13]** P. Ferrara: *"A generic static analyzer for multithreaded Java programs"*, Software: Practice and Experience, Volume 43, Issue 6, pages 663-684, June 2013 (ERA: A, AM Journal: 56th/376)

INTERNATIONAL CONFERENCES

- [BNO26]** G. Boldini, L. Negrini, L. Olivieri, P. Ferrara: “*A Modular Framework for Stack-Heap and Value Abstractions*”, 33rd Static Analysis Symposium (SAS 2026), Oakland, California, United States, October 3-9, 2026
- [CFL26]** G. Caiazza, P. Ferrara, T. Lisovenko, M. Biondo, D. Tosato: “*CoreTrust: Ensuring Data Quality at the Edge*”, 20th European Conference on Software Architecture (ECSA 2026), Bolzano, Italy, September 7-11, 2026
- [ANZ26]** V. Arceri, L. Negrini, G. Zanatta, F. Bianchi, T. Lisovenko, L. Olivieri, and P. Ferrara: “*JLiSA: The Java Frontend of the Library for Static Analysis (Competition Contribution)*”, 32nd International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2026), Turin, Italy, April 11-16, 2026 (CORE: A)
- [OBN25]** L. Olivieri, D. Beste, L. Negrini, L. Schönherr, A. E. Cinà, P. Ferrara: “*Code Generation of Smart Contracts with LLMs: A Case Study on Hyperledger Fabric*”, Proceedings of the 36th IEEE International Symposium on Software Reliability Engineering (ISSRE 2025), São Paulo, Brazil, October 21-24, 2025 (CORE: A)
- [OPN25]** L. Olivieri, L. Pasetto, L. Negrini, P. Ferrara: “*An Overview of Termination in the Ethereum Blockchain*”, Proceedings of the 4th EAI International Conference on Blockchain Technology and Emerging Applications (BlockTEA 2025), Venice, Italy, September 18-19, 2025
- [CLP25]** G. Caiazza, T. Lisovenko, P. Ferrara, F. Berti, F. Ferrari, A. Zaupa, G. Zhang: “*From Legacy to Intelligent IIoT Systems: Automation, Scalability and Elasticity*”, Proceedings of the IEEE 22nd International Conference on Software Architecture (ICSA 2025), Odense, Denmark, April 3-5, 2025 (CORE: A)
- [ONA25b]** L. Olivieri, L. Negrini, V. Arceri, P. Ferrara, A. Cortesi: “*Detection of Read-Write Issues in Hyperledger Fabric Smart Contracts*”, Proceedings of the 40th ACM Symposium on Applied Computing (track on Decentralized Applications - DAPP), Catania, Italy, March 31 – April 4, 2025
- [ONA25]** L. Olivieri, L. Negrini, V. Arceri, P. Ferrara, A. Cortesi, F. Spoto: “*Static Detection of Untrusted Cross-Contract Invocations in Go Smart Contracts*”, Proceedings of the 40th ACM Symposium on Applied Computing (track on Decentralized Applications - DAPP), Catania, Italy, March 31 – April 4, 2025
- [ZCF24]** G. Zanatta, G. Caiazza, P. Ferrara, L. Negrini, R. White: “*Automating ROS2 Security Policies Extraction through Static Analysis*”, Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024), Abu Dhabi, United Arab Emirates, October 14-18, 2024 (CORE: A)
- [OAN23]** L. Olivieri, V. Arceri, L. Negrini, F. Tagliaferro, P. Ferrara, A. Cortesi, F. Spoto: “*Flow Analysis for Detecting Non-Determinism in Blockchain*”, Proceedings of the 36th European Conference on Object-Oriented Programming (ECOOP 2023), LIPICs, Seattle, U.S.A., July 17-21, 2023 (CORE: A)
- [AOC22]** V. Arceri, M. Oliaro, A. Cortesi, P. Ferrara: “*Relational String Abstract Domains*”, Proceedings of the 23th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2022), LNCS, Springer, Philadelphia, U.S.A., January 16-18, 2022 (MS SE: 34th/285, AM PLSE: 8th/342)
- [NAF21]** Luca Negrini, Vincenzo Arceri, Pietro Ferrara, Agostino Cortesi: “*Twinning Automata and Regular Expressions for String Static Analysis*”, Proceedings of the 22th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2021), LNCS, Springer, Online, January 17-19, 2021 (MS SE: 34th/285, AM PLSE: 8th/342)
- [CPL20]** S. Calzavara, P. Ferrara, C. Lucchese: “*Certifying Decision Trees Against Evasion Attacks by Program Analysis*”, Proceedings of the 25th European Symposium on Research in Computer Security (ESORICS 20), LNCS, Springer, Guildford, United Kingdom, September 14-18, 2020 (CORE: A)
- [SCF20]** R. Salvia, A. Cortesi, P. Ferrara, F. Spoto: “*Intents Analysis of Android Apps for Confidentiality Leakage Detection*”, Advanced Computing and Systems for Security, 2020
- [FN20]** P. Ferrara, L. Negrini: “*SARL: OO Framework Specification for Static Analysis*”, Proceedings of the 12th Working Conference on Verified Software: Theories, Tools, and Experiments (VSTTE 2020), LNCS, Springer, Los Angeles, USA, July 20-21, 2020
- [MPK20]** A. Mandal, P. Ferrara, Y. Khlyebnikov, A. Cortesi, F. Spoto: “*Cross-Program Taint Analysis for IoT Systems*”, Proceedings of the 35th ACM Symposium on Applied Computing (SAC 2020), ACM Press, Brno, Czech Republic, March 30-April 3, 2020 (MS SE: 17th/285, AM PLSE: 43th/342)

- [FOS20]** P. Ferrara, L. Olivieri, F. Spoto: “*BackFlow: Backward Context-sensitive Flow Reconstruction of Taint Analysis Results*”, Proceedings of the 21th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2020), LNCS, Springer, New Orleans, United States, January 19-21, 2020 (MS SE: 34th/285, AM PLSE: 8th/342)
- [PCF18]** F. Panarotto, A. Cortesi, P. Ferrara, A. Mandal and F. Spoto: “*Static Analysis of Android Apps Interaction with Automotive CAN*”, Proceedings of the 3rd International Conference on Smart Computing and Communication (SmartCom 2018), Tokyo, Japan, December 10-12, 2018
- [SFS18]** R. Salvia, P. Ferrara, F. Spoto and A. Cortesi: “*Static Detection of Leaks across Intents*”, Proceedings of the 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications (TrustCom 2018), New York, USA, July 31-August 3, 2018 (CORE: A, MS CS 11th/793)
- [FOS18]** P. Ferrara, L. Olivieri and F. Spoto: “*Tailoring Taint Analysis to GDPR*”, Proceedings of the Annual Privacy Forum 2018, Barcelona, Spain, June 13-14, 2018
- [FCS18]** P. Ferrara, A. Cortesi and F. Spoto: “*CIL to Java-bytecode Translation for Static Analysis Leveraging*”, Proceedings of the 6th Conference on Formal Methods in Software Engineering (FormalISE 2018), Gothenburg, Sweden, June 2, 2018 (ACM SIGSOFT FME Distinguished Paper Award)
- [MCF18]** A. Mandal, A. Cortesi, P. Ferrara, F. Panarotto and F. Spoto: “*Vulnerability Analysis of Android Auto Infotainment Apps*”, Proceedings of the ACM International Conference on Computing Frontiers 2018 (CF 2018), Ischia, Italy, May 8-10, 2018
- [PPH17]** D. Piorkowski, S. Penney, A. Z. Henley, M. Pistoia, M. M. Burnett, O. Tripp, P. Ferrara: “*Foraging goes mobile: Foraging while debugging on mobile devices*”, Proceedings of IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2017), Raleigh, USA, October 11-14, 2017 (acceptance rate: 29%, Honorable mention)
- [APT17]** A. Aydin, D. Piorkowski, O. Tripp, P. Ferrara and M. Pistoia: “*Visual Configuration of Mobile Privacy Policies*”, Proceedings of the 20th International Conference on Fundamental Approaches to Software Engineering (FASE 2017), ARCoSS, LNCS, Springer, Uppsala, Sweden, April 22-29, 2017 (acceptance rate: 26%, MS SE: 29th/285, AM PLSE: 32th/342)
- [FTLK17]** P. Ferrara, O. Tripp, P. Liu, and E. Koskinen: “*Using Abstract Interpretation to Correct Synchronization Faults*”, Proceedings of the 18th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2017), ARCoSS, LNCS, Springer, Paris, France, January 15-17, 2017 (MS SE: 34th/285, AM PLSE: 8th/342)
- [TPT16]** P. Tsankov, M. Pistoia, O. Tripp, M. Vechev, and P. Ferrara: “*FASE: Functionality-Aware Security Enforcement*”, Proceedings of the 31st ACM Annual Computer Security Applications Conference (ACSAC 2015), ACM Press, Los Angeles, USA, December 7-11, 2015 (CORE: A, acceptance rate: 24%, MS SP 12th/138, AM SP: 17th/97)
- [FTP15]** P. Ferrara, O. Tripp, and M. Pistoia: “*MorphDroid: Fine-grained Privacy Verification*”, Proceedings of the 31st ACM Annual Computer Security Applications Conference (ACSAC 2015), ACM Press, Los Angeles, USA, December 7-11, 2015 (CORE: A, acceptance rate: 24%, MS SP 12th/138, AM SP: 17th/97)
- [BFTP15]** L. Brutschy, P. Ferrara, O. Tripp, and M. Pistoia: “*ShamDroid: Gracefully Degrading Functionality in the Presence of Limited Resource Access*”, Proceedings of the 30th ACM Conference on Object-oriented Programming (OOPSLA 2015), ACM Press, Pittsburgh, USA, October 27-30, 2015 (CORE: A*, acceptance rate: 25%, MS SE 4th/285, PL: 4th/168, AM PLSE: 5th/342)
- [BCF15]** G. Barbon, A. Cortesi, P. Ferrara, M. Pistoia and O. Tripp: “*Privacy Analysis of Android Apps: Implicit Flows and Quantitative Analysis (invited paper)*”, in Proceedings of the 14th International Conference on Computer Information Systems and Industrial Management Applications (CISIM 2015), LNCS, Springer, Warsaw, Poland, September 24-26, 2015
- [FMN15]** P. Ferrara, P. Müller and M. Novacek: “*Automatic Inference of Heap Properties Exploiting Value Domains*”, Proceedings of the 16th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2015), ARCoSS, LNCS, Springer, Mumbai, India, January 12-14, 2015 (MS SE: 34th/285, AM PLSE: 8th/342)
- [CFPT15]** A. Cortesi, P. Ferrara, M. Pistoia and O. Tripp: “*Datacentric Semantics for Verification of Privacy Policy Compliance by Mobile Applications*”, Proceedings of the 16th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2015), ARCoSS, LNCS, Springer, Mumbai, India, January 12-14, 2015 (MS SE: 34th/285, AM PLSE: 8th/342)

- [BFM14]** L. Brutschy, P. Ferrara, and P. Müller: “*Static Analysis for Independent App Developers*”, Proceedings of the 29th ACM Conference on Object-oriented Programming (OOPSLA 2014), ACM Press, Portland, USA, October 20-24, 2014 (CORE: A*, acceptance rate: 28%, MS SE 4th/285, PL: 4th/168, AM PLSE: 5th/342)
- [TFP14]** O. Tripp, P. Ferrara and M. Pistoia: “*Hybrid Security Analysis of Web JavaScript Code via Dynamic Partial Evaluation*”, Proceedings of the International Symposium on Software Testing and Analysis (ISSTA 2014), ACM Press, San Jose, USA, July 23-25, 2014 (CORE: A, acceptance rate: 28%, MS SE: 24th/285, AM PLSE: 4th/342, ACM SIGSOFT Distinguished Paper Award)
- [FSB14]** P. Ferrara, D. Schweizer and L. Brutschy: “*TouchCost: Cost Analysis of TouchDevelop Scripts*”, Proceedings of the 17th International Conference on Fundamental Approaches to Software Engineering (FASE 2014), ARCoSS, LNCS, Springer, Grenoble, France, April 5-13, 2014 (acceptance rate: 26%, MS SE: 29th/285, AM PLSE: 32th/342)
- [Ferr14]** P. Ferrara: “*Generic Combination of Heap and Value Analyses in Abstract Interpretation*”, Proceedings of the 15th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2014), ARCoSS, LNCS, Springer, San Diego, USA, January 19-21, 2014 (MS SE: 34th/285, AM PLSE: 8th/342)
- [CFCM13]** G. Costantini, P. Ferrara, A. Cortesi and G. Maggiore: “*The Domain of Parametric Hypercubes for Static Analysis of Computer Games Software*”, Proceedings of the 15th International Conference on Formal Engineering Methods (ICFEM 2013), LNCS, Springer, Queenstown, New Zealand, October 29-November 1, 2013 (acceptance rate: 32%)
- [CFC13]** A. Cortesi, P. Ferrara and N. Chaki: “*Static Analysis Techniques for Robotics Software Verification (invited paper)*”, in Proceedings of the 44th International Symposium of Robotics (ISR 2013), Seoul, Korea, October 24-26, 2013 (CORE: A)
- [CFC12]** G. Costantini, P. Ferrara and A. Cortesi: “*Linear approximation of continuous systems with Trapezoid Step Functions*”, Proceedings of the 10th Asian Symposium on Programming Languages and Systems (APLAS 2012), LNCS, Springer, Kyoto, Japan, December 11-13, 2012 (MS PL: 42th/168, AM PLSE: 80th/342)
- [FFJ12]** P. Ferrara, R. Fuchs and U. Juhasz: “*TVAL+: TVLA and Value Analyses Together*”, Proceedings of the 10th International Conference on Software Engineering and Formal Methods (SEFM 2012), LNCS, Springer, Thessaloniki, Greece, October 3-5, 2012 (acceptance rate: 25%, MS SE: 76th/285, AM PLSE: 104th/342)
- [ZFC12]** M. Zanioli, P. Ferrara and A. Cortesi: “*SAILS: static analysis of information leakage with Sample*”, Proceedings of the 27th ACM Symposium on Applied Computing (SAC 2012), ACM Press, Riva del Garda, Italy, March 26-30, 2012 (acceptance rate: 26%, MS SE: 17th/285, AM PLSE: 43th/342)
- [FM12]** P. Ferrara and P. Müller: “*Automatic inference of access permissions*”, Proceedings of the 13th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2012), LNCS, Springer, Philadelphia, USA, January 22-24, 2012 (MS SE: 34th/285, AM PLSE: 8th/342)
- [CFC11]** G. Costantini, P. Ferrara and A. Cortesi: “*Static analysis of string values*”, Proceedings of the 13th International Conference on Formal Engineering Methods (ICFEM 2011), LNCS, Springer, Durham, United Kingdom, October 25-28, 2011
- [Ferr10]** P. Ferrara: “*Static type analysis of pattern matching by abstract interpretation*”, Proceedings of the IFIP Conference on Formal Techniques for Distributed Systems (FORTE/FMOODS 2010), LNCS, Springer, Amsterdam, Netherlands, June 7-10, 2010 (CORE: A, MS DPC: 45th/203, AM NET: 18th/177)
- [Ferr09a]** P. Ferrara: “*Checkmate: a generic static analyzer of Java multithreaded programs*”, Proceedings of the 7th IEEE International Conference on Software Engineering and Formal Methods (SEFM 2009), IEEE Computer Society, Hanoi, Vietnam, November 23-27, 2009 (acceptance rate: 35%, MS SE: 76th/285, AM PLSE: 104th/342)
- [Ferr08b]** P. Ferrara: “*Static analysis of the determinism of multithreaded programs*”, Proceedings of the 6th IEEE International Conference on Software Engineering and Formal Methods (SEFM 2008), IEEE Computer Society, Cape Town, South Africa, November 10-14, 2008 (CORE: A, acceptance rate: 28%, MS SE: 76th/285, AM PLSE: 104th/342)
- [FLF08]** P. Ferrara, F. Logozzo and M. Fähndrich: “*Safer unsafe code for .NET*”, Proceedings of the 23rd ACM Conference on Object-oriented Programming (OOPSLA 2008), ACM Press, Nashville, USA, October 19-23, 2008 (CORE: A*, acceptance rate: 28%, MS SE: 4th/285, MS PL: 4th/168, AM PLSE: 5th/342)
- [Ferr08]** P. Ferrara: “*Static analysis via abstract interpretation of the happens-before memory model*”, Proceedings of the 2nd International Conference on Tests and Proofs (TAP 2008), LNCS, vol. 4966, Springer, Prato, Italy, April 9-11, 2008

INTERNATIONAL WORKSHOPS WITH PROGRAM COMMITTEE

- [CCB25]** B. Chachar, M. Cavazza, A. Bracciali, P. Ferrara, and A. Cortesi: *"Finding Vulnerabilities in Solidity Smart Contracts with In-Context Learning"*, Proceedings of the Joint National Conference on Cybersecurity (ITASEC & SERICS 2025), Bologna, Italy, February 3-8, 2025
- [OPN24]** L. Olivieri, L. Pasetto, L. Negrini, P. Ferrara: *"European Union Data Act and Blockchain Technology: Challenges and New Directions"*, 6th Distributed Ledger Technologies Workshop (DLT2024), Turin, Italy, May 14-15, 2024
- [NPF24]** L. Negrini, S. Presotto, P. Ferrara, E. Zaffanella, A. Cortesi: *"Stability: an abstract domain for the trend of variation of numerical variables"*, ACM SIGPLAN International Workshop on Numerical and Symbolic Abstract Domains (NSAD 2024), Pasadena CA, USA, October 22, 2024
- [ZFL24]** G. Zanatta, P. Ferrara, T. Lisovenko, L. Negrini, G. Caiazza, R. White: *"Sound Static Analysis for Microservices: Utopia? A preliminary experience with LiSA"*, Formal Techniques for Java-like Programs (FTfJP 2024), Wien, Austria, September 20, 2024
- [NAO24]** L. Negrini, V. Arceri, L. Olivieri, A. Cortesi, P. Ferrara: *"Teaching through Practice: Advanced Static Analysis with LiSA"*, Formal Methods Teaching Workshop (FMTea 2024), Milan, Italy, September 10, 2024
- [OPN24]** L. Olivieri, L. Pasetto, L. Negrini, P. Ferrara: *"European Union Data Act and Blockchain Technology: Challenges and New Directions"*, 6th Distributed Ledger Technologies Workshop (DLT 2024), Turin, Italy, May 14-15, 2024
- [Ferr23]** P. Ferrara: *"[Micro]services: threat, challenge, or opportunity for sound static program analysis?"*, Fifth International Conference on Microservices (Microservices 2023), Pisa, Italy, October 10-11, 2023
- [OTA22]** L. Olivieri, F. Tagliaferro, V. Arceri, M. Ruaro, L. Negrini, A. Cortesi, P. Ferrara, F. Spoto, E. Talin: *"Ensuring Determinism in Blockchain Software with GoLiSA: An Industrial Experience Report"*, 11th ACM SIGPLAN International Workshop on the State of the Art in Program Analysis (SOAP 2022), San Diego, California, United States, June 14, 2022
- [FNA21]** P. Ferrara, L. Negrini, V. Arceri, A. Cortesi: *"Static Analysis for Dummies: Experiencing LiSA"*, 10th ACM SIGPLAN International Workshop on the State of the Art in Program Analysis (SOAP 2021), Online, June 22, 2021
- [FMCa19]** P. Ferrara, A. Mandal, A. Cortesi, F. Spoto: *"Static Analysis for the OWASP IoT Top 10 2018"*, Workshop on Security practices for Internet of Things (SPIoT 2019), Prague, Czech Republic, April 7, 2019
- [FMC19]** P. Ferrara, A. Mandal, A. Cortesi, F. Spoto: *"Cross Programming Language Taint Analysis for the IoT Ecosystem"*, Interactive Workshop on the Industrial Application of Verification and Testing (InterAVT 2019), Prague, Czech Republic, April 6, 2019
- [FA19]** P. Ferrara, P. Anderson: *"Semantic Static Analysis of IoT Software"*, Proceedings of Embedded World 2019, Nuremberg, Germany, February 26-28, 2019
- [NF18]** L. Negrini, P. Ferrara: *"SARL: Framework Specification for Static Analysis"*, Proceedings of The Ninth Workshop on Tools for Automatic Program Analysis (TAPAS 2018), Freiburg im Breisgau, Germany, August 28, 2018
- [FS18]** P. Ferrara and F. Spoto: *"Static Analysis for GDPR Compliance"*, Proceedings of the 2nd Italian Conference on Cybersecurity (ITASEC 2018), Milan, Italy, February 6-8, 2018
- [BFS17]** E. Burato, P. Ferrara and F. Spoto: *"Security Analysis of the OWASP Benchmark with Julia"*, Proceedings of the 1st Italian Conference on Cybersecurity (ITASEC 2017), Venice, Italy, January 17-20, 2017
- [BSCF16]** Gianluca Barbon, Enrico Steffinlongo, Agostino Cortesi and Pietro Ferrara: *"DAPA: Degradation-Aware Privacy Analysis of Android Apps"*, Proceedings of the 12th International Workshop on Security and Trust Management (STM 2016), Heraklion, Greece, September 26-27, 2016
- [TPFR16]** Omer Tripp, Marco Pistoia, Pietro Ferrara, and Julia Rubin: *"Pinpointing mobile malware using code analysis"*, Proceedings of the IEEE/ACM International Conference on Mobile Software Engineering and Systems (MOBILESoft 2016), Austin, USA, May 16-17 2016

- [PTF15]** Marco Pistoia, Omer Tripp, Pietro Ferrara, and Paolina Centonze: *“Automatic Detection, Correction and Visualization of Security Vulnerabilities in Mobile Apps (invited paper)”*, Proceedings of the 3rd Workshop on Mobile Development Lifecycle (MobileDeLi 2015), Pittsburgh, USA, October 26, 2015
- [BFM14a]** L. Brutschy, P. Ferrara, and P. Müller: *“TouchGuru: Integrating Static Analysis with a Mobile Development Environment (invited paper)”*, Proceedings of the 2nd Workshop on Mobile Development Lifecycle (MobileDeLi 2014), Portland, USA, October 21, 2014
- [CCF13]** A. Cortesi, G. Costantini and P. Ferrara: *“A survey on Product Operators in Abstract Interpretation”*, Proceedings of the Festschrift for Dave Schmidt, Electronic Proceedings in Theoretical Computer Science, U.S.A., September 19-20, 2013
- [Ferr08a]** P. Ferrara: *“A fast and precise analysis for data race detection”*, Proceedings of the Third Workshop on Bytecode Semantics, Verification, Analysis and Transformation (Bytecode 2008), Budapest, Hungary, April 6, 2008
- [Ferr06]** P. Ferrara: *“JAIL: Firewall Analysis of Java Card by Abstract Interpretation”*, Proceedings of the 1st International Workshop on Emerging Applications of Abstract Interpretation (EAAI 2006), Vienna, Austria, March 26, 2006

THESIS

- [Ferr09]** P. Ferrara: *“Static analysis via abstract interpretation of multithread programs”*, PhD thesis under the supervision of Radhia Cousot and Agostino Cortesi defended at Ecole Normale Supérieure of Paris on May 22nd, 2009

For each conference and journal publication, the following metrics are reported:

- acceptance rate when less than 35%
- CORE: the CORE ranking¹ when A* or A (all others are B)
- ERA: the ERA 2010 journal ranking²
- MS: Microsoft Academics ranking³ when in the top 30% of conferences/journals in:
 - Software Engineering (SE)
 - Security & Privacy (SP)
 - Computer Security (CS)
 - Programming Languages (PL)
 - Distributed & Parallel Computing (DPC)
 - Journals in Computer Science (Journal)
- AM: Arnet Miner ranking⁴ when in the top 30% of conferences in:
 - Programming Languages and Software Engineering (PLSE)
 - Network (NET)
 - Security and Privacy (SP)

Community Service

¹<http://103.1.187.206/core/>

²https://research.unsw.edu.au/sites/all/files/related_files/regular_page_content/era2010_journal_title_list.xls

³<http://academic.research.microsoft.com/?SearchDomain=2&entitytype=3>

⁴<http://arnetminer.org/page/conference-rank/html/All-in-one.html>

2027 **Review Committee member**, OOPSLA

July 2026 **Opponent**, PhD committee of Dario Amoroso D’Aragona at Tampere University, Finland

2026 **Program Committee member**, ISSTA

2026 **Doctoral Symposium Co-Chair**, ECOOP

2025 **Financial Chair**, IEEE Euro S&P

From 2022 **Program Co-Chair, General Chair**, CSV

2025 **Review Committee member**, OOPSLA

December 2025 **Reviewer**, PhD committee of Charles Babu at Université de Lorraine, France

October 2025 **Organizer**, Dagstuhl Seminar 25421 “Sound Static Program Analysis in Modern Software Engineering”

June 2025 **Reviewer**, PhD committee of Michael Schwarz at Technische Universität München, Germany

2024, 2025 **Program Committee member**, BuildSEC - Building a Secure & Empowered Cyberspace

2024 **Guest editor**, Special issue of International Journal on Software Tools for Technology Transfer (ISSN 1433-2779, Springer) on CSV 2024

2023 **Program Co-Chair**, SOAP

2023 **Program Committee member**, SAS

2023 **Guest editor**, Special issue of International Journal on Software Tools for Technology Transfer (ISSN 1433-2779, Springer) on SOAP 2023

May 2023 **Scientific Board**, ESA_Labs Annual Event: Space Data Management Workshop

July 2023 **Invited Participant**, Dagstuhl Seminar 20251 “Theoretical Advances and Emerging Applications in Abstract Interpretation”

2023 **Guest editor**, Special chapter of International Journal on Software Tools for Technology Transfer (ISSN 1433-2779, Springer) on CSV 2023

2024 **External reviewer of a grant proposal**, Deutsche Forschungsgemeinschaft (German Research Foundation)

2023 **Program Committee member**, 16th International Conference on Knowledge Science, Engineering and Management (KSEM)

2023 **Guest editor**, Proceedings of “Challenges of Software Verification 2022”, “Intelligent Systems Reference Library” book series, Springer-Nature. <https://link.springer.com/book/9789811996009>

Guest editor, Special Issue “Validation Methods in IoT Systems: Security, Performance and Safety” of Sensors (ISSN 1424-8220, MDPI). https://www.mdpi.com/journal/sensors/special_issues/IoT_Safety

2022 **Program Committee member**, SAS

2022 **Artifact Evaluation Committee member**, PLDI

2022 **Program Committee member**, SOAP

2022 **Program Committee member**, 15th International Conference on Knowledge Science, Engineering and Management (KSEM)

From 01/2021 **Review Editor**, Frontiers in Computer Science (Computer Security)

2020 **Program Committee member**, 9th International Workshop on Numerical and Symbolic Abstract Domains (NSAD 2020)

2020 **Program Committee member**, 12th Working Conference on Verified Software: Theories, Tools, and Experiments (VSTTE)

Guest editor, Special issue of the journal Applied Sciences (ISSN 2076-3417, section “Computing and Artificial Intelligence”, MDPI) on “Static Analysis Techniques: Recent Advances and New Horizons”. https://www.mdpi.com/journal/applsci/special_issues/Static_Analysis_Techniques

2019 **Program Committee member**, SmartCom

2018 **Program Committee member**, SOAP

2017 **Artifact Evaluation Committee member**, CAV

2016 **Program Committee member**, FTfJP

2015 **Demonstrations track co-chair**, SPLASH

2015 **Organizer**, IBM Programming Languages Day

Apr. 2015 **External reader**, PhD committee of Junjie Chen at the New York University

2014 **External reviewer of a grant proposal**, Swiss National Science Foundation

2012 to 2023 **Program Committee member**, CISIM

2012 **Program Committee member, Session chair**, SAS

Reviewer for the following conferences, workshops and journals: SAS 07, FTfJP 07, VMCAI 08, ESOP 08, LCTES 08, FM 09, CATS 2010, ESOP 2010, Journal of Symbolic Computation – Special issue on WING 2009, FMOODS 2010, AP-NOC 2010, FoVeOOS 2010, ESOP 2011, Journal of Symbolic Computation – Special issue on WING 2010, TOPI 2011, PSI 2011, SAS 2011, ESOP 2012, TACAS 2012, Bytecode 2012, Journal of Computer Science and Technology, Journal of Software and Systems Modeling, VMCAI 2013, ESOP 2013, ICFEM 2013, POPL 2014, Formal Aspects of Computing, CAV 2014, APLAS 2014, VMCAI 2015, NFM 2015, Science of Computer Programming, Computer Languages Systems & Structures – Special issue on VMCAI 2015, VMCAI 2016, ICALP 2016, ESOP 2017, PLDI 2017, Security and Communication Networks, SAS 2017, Computer Languages, Systems & Structures, Journal of Systems and Software, Computer Networks, Sustainable Cities and Society, Security and Communication Networks, Journal of King Saud University - Computer and Information Sciences, Software Tools for Technology Transfer - Special issue on SPIoT 2019, Journal of Logical and Algebraic Methods in Programming, MDPI Computer (Special Issue: Integration of Cloud Computing and IoT), International Journal of Security and Networks, IEEE Transactions on Software Engineering, Formal Methods in System Design, ACM Transactions on Software Engineering and Methodology, IEEE Transactions on Network Science and Engineering, Information and Software Technology.

Grants

Faculty Participant, "SUPREME: Sistemi Umano centrati per Prodotti e pRocEssi Manufatturieri Evoluti"

EUR 150'000, Project 3,9MEUR

REGIONE VENETO, RETI INNOVATIVE REGIONALI

January 2025 - December 2026

The main goal of the project is to design Industry 5.0 systems to monitor and optimize industrial plants in terms of reliability, energy efficiency, and quality control.

Faculty Participant, "SATCO: Satellite Tattico Cognitivo"

EUR 95'000, Project 2MEUR

REGIONE VENETO, RETI INNOVATIVE REGIONALI

January 2025 - December 2026

The main goal of the project is to develop a constellation of satellites offering on-demand optical and laser signal intelligence services.

Principal Investigator, "Digital platform 5.0 transformation"

EUR 40'000

SMACCT COMPETENCE CENTER

March 2025 - March 2026

The main goal of this industrial project is to develop a scalable and elastic software architecture to collect and elaborate data coming from rollercoasters. The project is in collaboration with Zamperla SPA.

Member of the research unit at Ca' Foscari University of Venice

87KEUR, project 400KEUR

EUROPEAN UNION, ERASMUS+

September 2024 - June 2027

EcoDigify aims to initiate a future-oriented interdisciplinary university program that will address the multidimensional challenges of promoting sustainability in the use and development of digital technologies. This will be achieved by enhancing the educators' knowledge about Sustainable Digitalization and providing students with the knowledge and skills to co-create sustainable digital solutions with the support of digitalization experts and sustainability scientists.

Member of the research unit at Ca' Foscari University of Venice

Spoke 6 6,3MEUR, Project 114MEUR

ITALIAN MINISTRY OF UNIVERSITY AND RESEARCH, PARTENARIATI ESTESI, SERICS - SECURITY AND RIGHTS IN THE CYBERSPACE

January 2023 - December 2025

Spoke 6 (Software and System Security). The project focuses on various methodologies to identify, detect, prevent, and fix security vulnerabilities in software. In particular, my main research focus was on the activities of SWOPS (Securing softWare frOm first Principles) and SCAI (Supply Chain Attack Avoidance).

Task leader, "Dynamic modeling and trajectory planning for mechatronic and robotic systems, AI-driven robots, specification and validation of functional and security requirements of robotic software"

Spoke 3, 7MEUR

ITALIAN MINISTRY OF UNIVERSITY AND RESEARCH, ECOSISTEMI DELL'INNOVAZIONE, INEST - INTERCONNECTED NORD-EST

INNOVATION ECOSYSTEM

September 2022 - September 2025

Spoke 3, Research Topic 2 "Smart Manufacturing, Mechatronics and Robotics", Task S3_RT2.4. A survey on skills and expertise of Italian north-east industrial sector will allow the identification of possible topics to be investigated exploiting the synergies between industry and research institutions. A review on the state of the art on dynamic modeling and trajectory planning for mechatronics and robotic systems, AI-driven robots and security requirements of robotic software will be conceived exploiting the expertise of each participant. Ultimately, the analysis of performance and key performance indicators of implementations will help shaping the digital transformation, extending the benefits of digital technologies to the main areas of specialization in the industrial-manufacturing, agriculture, health and food sectors.

Principal Investigator, "Static Analysis for Data Scientists"

EUR 48'000

CA' FOSCARI UNIVERSITY OF VENICE, SPIN PROJECT

October 2021 - October 2023

Big data revolutionized the world of software development. The novel figure of data scientists arised in the job market. Those are not expert software developers, and they often obtained bachelor or master degrees in different scientific fields (e.g., in statistics, physics, or mathematics). The main goal of this project is to develop an effective tool based on static analysis to help data scientists developing Python scripts for data processing purposes.

Principal Investigator, "IAM access control policies verification and inference"

USD 80'000

AMAZON RESEARCH AWARDS, AWS AUTOMATED REASONING

June 2021 - June 2022

The goal of this project is to apply abstract interpretation techniques to over approximate the string values computed by applications using AWS cloud services, and then infer, validate or IAM the access control policies exploiting such string information.

Co-principal Investigator, "Families-to-Families for sharing children care"

EUR 41'400

FONDO INTEGRATIVO SPECIALE PER LA RICERCA FISIR - COVID

June 2021 - December 2021

The objective is to specialize the Families_Share (a previous H2020 project) model to the specific needs related to post-lockdown and to provide the tools for activating capillary help-aid groups throughout the country.

Faculty Participant, "VIR2EM: Virtualization and Remotization for Resilient and Efficient Manufacturing"

EUR 140'000

REGIONE VENETO - POR FESR 2014-2020, RETI INNOVATIVE REGIONALI

September 2020 - September 2022

The main goal is to develop a variety of static analyzer to perform security assessments of Industry 4.0 software, as well as a dashboard that allows non-expert users to take actions to improve the security of such systems.

Fondo di primo insediamento (First settlement fund)

EUR 18'000

CA' FOSCARI UNIVERSITY OF VENICE

May 2020 - May 2022

The goal of this project is to (i) develop a generic framework to formalize and approximate the communication channels adopted by different IoT layers to communicate, (ii) apply such abstraction to some of the most popular programming languages adopted for developing IoT software (Java, Python, JavaScript) relying on existing generic static analyzers, and (iii) instrument taint analysis considering exactly what portions of the communications between different IoT software layers are tainted detecting potential security vulnerabilities.

Co-principal investigator, "Development and verification of e-commerce blockchain systems"

EUR 99'000

REGIONE VENETO ESF PROJECT

Jul. 2020 - Sept. 2021

The goal of this project is to develop a blockchain application for the management of invoices, and a verifier based on static analysis to validate such application. The project is 1-year-long and involves two universities (Venice and Verona) and three enterprises (JuliaSoft, Commerc.io, and Alpenite). Three postdocs are involved in the project.

Co-principal investigator, "Security Static Analysis for Android Things"

EUR 165'000

UNIVERSITY OF VERONA

Oct. 2018 - Nov. 2019

The goal of this project is to formalize, implement and apply novel security analyses for Android Things software. The project is 2-years-long and involves one postdoc and two professors at University of Verona and two employees at Julia.

Co-principal investigator, "Static analysis for the safety and security of Android systems for automotive infotainment"

EUR 48'000

REGIONE VENETO ESF PROJECT

Jul. 2017 - Jul. 2018

The goal of this project is to formalize, implement and apply novel reliability and security analyses for Android Auto apps. The project is 1-year-long and involves two postdocs (one at University Ca' Foscari of Venice, and one at University of Verona that will spend at least 7 months at Julia).

Co-principal investigator, "TouchBoost – Cloud-based Static Analyses to Improve Lay Programming on Mobile Devices"

CHF 298'000

MICROSOFT

Apr. 2012 - Jul. 2013

The goal of this project is to apply static analysis to improve the efficiency and reliability of programs developed on a mobile device with Microsoft TouchDevelop. The project is 2-years-long and involves one postdoc and one PhD student (Lucas Brutschy).

Technical head, "Verification-Driven Inference of Contracts"

CHF 302'896

SWISS NATIONAL FOUNDATION

Sep. 2010 - Jul. 2013

Project leader: prof. Peter Müller. The goal of this project is to explore the interaction between program verification and static analysis in order to effectively find bugs on software. The project is 3-years-long and involves two PhD students (Uri Juhasz and Milos Novacek).

Institutional Responsibilities

2025	President , Commission of the state exam for the admission to the Italian register of computer engineers	<i>Ca' Foscari University, Venice, Italy</i>
From January 2024	Member , 3rd mission committee, Department of Environmental Sciences, Informatics and Statistics	<i>Ca' Foscari University, Venice, Italy</i>
39th cycle	Member , Examination board, PhD in Computer Science	<i>Ca' Foscari University, Venice, Italy</i>
From 2023	Co-Organizer with Andrea Marin , Google DevFest Venezia, 1 day-event, 4 parallel track, around 15 partner enterprises and 400 attendees	<i>Ca' Foscari University, Venice, Italy</i>
From January 2023	Member , Quality assurance group of the master in Computer Science and Information Technology (CM90)	<i>Ca' Foscari University, Venice, Italy</i>
From March 2021	Elected Member , Coordinamento interdipartimentale dei Ricercatori di Ca' Foscari	<i>Ca' Foscari University, Venice, Italy</i>
From May 2021	Coordinator with prof. Francesco Casarin , Gruppo scacchistico cafoscarino ⁵	<i>Ca' Foscari University, Venice, Italy</i>
From November 2020	Member , Teaching committee, Bachelor Degree Programme in Digital Management	<i>Ca' Foscari University, Venice, Italy</i>
36th cycle	Member , Examination board, PhD in Computer Science	<i>Ca' Foscari University, Venice, Italy</i>
From 29/10/2019	Member , Teaching Committee, PhD in Computer Science	<i>Ca' Foscari University, Venice, Italy</i>

Internships

Microsoft Research

INTERN

"Static analysis of unsafe code in C# via abstract interpretation", mentor: Francesco Logozzo.

Redmond - WA, U.S.A.

Aug. 2007 - Nov. 2007

Ca' Foscari University

COLLABORATOR

"Development of a static analyzer for Java Card language"

Venice, Italy

Apr. 2005 - Jun. 2005

Ca' Foscari University

INTERN

I developed an application to evaluate automatically the practice exams of the "Base of programming" course and I help some students to develop simple C applications in Linux environment.

Venice, Italy

Sept. 2004 - Dec. 2004

Tools and Software

LiSA (Library for Static Analysis)

CA' FOSCARI UNIVERSITY

LiSA (Library for Static Analysis - <https://github.com/Unive-SSV/lisa>) aims to ease the creation and implementation of static analyzers based on the Abstract Interpretation theory. LiSA provides an analysis engine that works on a generic and extensible control flow graph representation of the program to analyze. Abstract interpreters in LiSA are built for analyzing such representation, providing a unique analysis infrastructure for all the analyzers that will rely on it. The development of LiSA is led by Luca Negrini, and the tool is part of several funded projects and used for teaching activities and exams.

Venice, Italy

from Sept. 2020

Julia for .NET

JULIASOFT

Apply Julia (a 10 years old static analyzer for Java) to the analysis of CIL (.Net bytecode) obtained from the compilation of C# code. Julia has been widely applied to the analysis of Java bytecode. The main challenge is to translate CIL (and in particular, instructions accessing the memory through pointers) to Java Bytecode (that does not support pointers). In particular, CIL is translated into an intermediate XML representation similar to Java bytecode, and this is passed to Julia as BCEL classes.

Verona, Italy

Feb. 2016 - Nov. 2016

⁵<https://www.unive.it/pag/42231/>

Sample: Static Analyzer of Multiple Programming Languages

Switzerland

ETH ZÜRICH

Oct. 2010 - Jul. 2013

A generic static analyzer that can be automatically plugged with a wide range of different heap and value analyses [Ferr14]. Several value analyses have been already developed [Ferr10, CFC11, ZFC12, FM12, CFC12, CFC14]. In addition, Sample supports several classical heap (e.g., TVLA [FFJ12] and program point-based abstraction) and numerical (e.g., Apron) analyses, as well as trace partitioning. Sample works on an intermediate language (called Simple) simpler than common programming languages, but more refined than bytecode languages. Up to now, translations from TouchDevelop, Scala and Java bytecode to Simple has been developed.

Checkmate: Static Analysis of Java Multithreaded Programs [Ferr09a,Ferr13]

Paris, France

ECOLE POLYTECHNIQUE

Apr. 2008 - Jan. 2010

A generic static analyzer of Java multithreaded programs developed in Java. It is generic w.r.t. the numerical domain, the memory model and the property. It has been instantiated on the analysis of the happens-before memory model [Ferr08], and of the determinism [Ferr08b], together with other more approximated memory models, properties, and a set of well-known non-relational numerical domains.

Static Analysis of Unsafe Code [FLF08]

Redmond – WA, U.S.A.

MICROSOFT RESEARCH

Aug. 2007 - Nov. 2007

An extension of Clousot (a generic static analyzer of MSIL programs) written in C# in order to analyze unsafe code (MSIL code dealing with pointers).

JAIL: Firewall Analysis of Java Card by Abstract Interpretation [Ferr06]

Venice, Italy

CA' FOSCARI UNIVERSITY

Sept. 2004 - Jun. 2005

Tool developed during my master's thesis and the following internship in Java. It analyzes Java Card source code and checks if the applet isolation property is respected.

Patents

2022

1. M Pistoia, O Tripp, P Ferrara, P Tsankov: "Value based information tracking for security enforcement", US Patent 11,244,057
2. P Ferrara, M Pistoia, O Tripp: "System, method and apparatus for extracting usage-based fine grained permissions", US Patent 11,227,045
3. O Tripp, JT Dolby, M Pistoia, P Ferrara: "User device privacy protection", US Patent 11,222,135

2021

4. L Brutschy, P Ferrara, M Pistoia, O Tripp: "Fine-grained user control over usages of sensitive system resources having private data with applications in privacy enforcement", US Patent 11,176,267
5. P Ferrara, M Pistoia, O Tripp, P Tsankov: "Light-weight context tracking and repair for preventing integrity and confidentiality violations", US Patent 11,138,319
6. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Privacy detection of a mobile application program", US Patent 10,915,659
7. Dietsch, Matthias Daniel; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Identifying computer program security access control violations using static analysis", US Patent 10,956,580

2020

8. Tripp, Omer; Pistoia, Marco; Ferrara, Pietro: "Cooperative updating of software", US Patent 10,846,080
9. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "System, method and apparatus for usable code-level statistical analysis with applications in malware detection", US Patent 10,846,401
10. Tripp, Omer; Ferrara, Pietro; Lubensky, David; Pistoia, Marco: "Data-centric approach to analysis", US Patent 10,838,915
11. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer; Tsankov, Petar I: "Providing efficient information tracking with dynamically selected precision", US Patent 10,701,099
12. Dolby, Julian T; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Install-Time Security Analysis of Mobile Applications", US Patent 10,621,333

2019

13. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Performing entropy-based dataflow analysis", US Patent 10,289,540
14. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "System, method and apparatus for fine-grained privacy specification and verification", US Patent 10,250,642

15. Bello, Luciano; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Security enforcement in the presence of dynamic code loading", US Patent 10,296,737

2018

16. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Semantic Privacy Enforcement", US Patent App. 15/223,804
17. Ferrara, Pietro; Pistoia, Marco; Ponzio, John; Tripp, Omer: "Application integrity verification in multi-tier architectures", US Patent 10,073,975
18. Ferrara, Pietro; Pistoia, Marco; Roos, Pascal; Tripp, Omer: "Machine learning to facilitate incremental static program analysis", US Patent App. 15/244,229
19. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Cooperative creation of dataflow models using sparse random instrumentation", US Patent 9,996,324

2017

20. Brutschy, Lucas; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Synthesizing inputs to preserve functionality", US Patent 9,697,018
21. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer; Tsankov, Petar I: "Self-repair and distributed-repair of applications", US Patent 9,684,788
22. Ferrara, Pietro; Pistoia, Marco; Tripp, Omer; Eunho, YANG: "Trace recovery via statistical reasoning", US Patent 9,823,998
23. Dolby, Julian T; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "Detection of software or hardware incompatibilities in software packages", US Patent 9,733,927
24. Bello, Luciano; Ferrara, Pietro; Pistoia, Marco; Tripp, Omer: "System and method for bypassing evasion tests with applications in analysis and monitoring of mobile applications", US Patent App. 15/166,817

Talks

INTERNATIONAL CONFERENCES AND WORKSHOPS

- | | |
|-----------------|--|
| 17/07/26 | JLiSA: from zero to podium at SV-COMP and Anecdotes, Helmutfest, Munich, Germany |
| 03/04/25 | From Legacy to Intelligent IIoT Systems: Automation, Scalability and Elasticity, ICSA 2025, Odense, Denmark |
| 11/10/23 | [Micro]services: threat, challenge, or opportunity for sound static program analysis?, Microservices 2023, Pisa, Italy |
| 14/07/23 | Teaching AI with LiSA, Dagstuhl-Seminar 23281 "Theoretical Advances and Emerging Applications in Abstract Interpretation", Germany |
| 31/03/20 | Cross-Program Taint Analysis for IoT Systems, SAC 2020 (SVT track), Brno, Czech Republic (on-line format) |
| 13/06/18 | Tailoring taint analysis for GDPR, Annual Privacy Forum 2018, Barcelona, Spain |
| 02/06/18 | CIL to Java-bytecode Translation for Static Analysis Leveraging, Formalise 2018, Gothenburg, Sweden |
| 08/02/18 | Static Analysis for GDPR Compliance, ITASEC 18, Milan, Italy |
| 06/02/18 | Code Analysis Reinvented: boosting software security and privacy (tutorial), ITASEC 18, Milan, Italy |
| 09/04/14 | TouchCost: Cost Analysis of TouchDevelop Scripts, FASE 14, Grenoble, France |
| 22/01/14 | Generic Combination of Heap and Value Analyses in Abstract Interpretation, VMCAI 14, San Diego, CA, USA |
| 11/12/12 | Linear approximation of continuous systems with Trapezoid Step Functions, APLAS 12, Kyoto, Japan |
| 03/10/12 | TVLA+: TVLA and Value Analyses Together, SEFM 12, Thessaloniki, Greece |
| 22/01/12 | Automatic inference of access permissions, VMCAI 12, Philadelphia, PA, USA |
| 26/10/11 | Static analysis of string values, ICFEM 11, Durham, Great Britain |
| 09/06/10 | Static type analysis of pattern matching by abstract interpretation, FORTE/FMOODS 10, Amsterdam, Netherlands |
| 26/11/09 | Checkmate: a Generic Static Analyzer of Java Multithreaded Programs, SEFM 09, Hanoi, Vietnam |

- 11/11/08** Static analysis of the determinism of multithreaded programs, SEFM 08, Cape Town, South Africa
- 10/04/08** Static analysis via abstract interpretation of the happens-before memory model, TAP 2008, Prato, Italy
- 05/04/08** A fast and precise alias analysis for data race detection, Bytecode 08 (ETAPS workshop), Budapest, Hungary
- 26/03/06** JAIL: Firewall Analysis of Java Card by Abstract Interpretation, EAAI '06 (ETAPS workshop), Wien, Austria

ACADEMIC INSTITUTIONS AND RESEARCH LABS

- 03/04/25** Sound Static Analysis of Modern Software Architecture, SDU, Odense, Denmark
- 01/04/25** Sound Static Analysis of Modern Software Architecture, Università di Bologna, Italy
- 20/03/25** Sound Static Analysis of Modern Software Architecture, SyCoMoRES, INRIA Lille, France
- 19/03/25** Sound Static Analysis of Modern Software Architecture, IRISA, INRIA Rennes, France
- 18/03/25** Sound Static Analysis of Modern Software Architecture, LIP6, Sorbonne University, Paris, France
- 11/06/18** The Julia static analyzer and its application to security vulnerabilities, “Sicurezza e affidabilità” course”, University Bicocca of Milan, Italy
- 26/05/18** Java and its bytecode, Android, static analysis, CyberChallenge.IT course, University Ca’ Foscari of Venice, Italy
- 26/03/18** La vulnerabilità del software nell’ecosistema IoT, seminar on cybersecurity in Industry 4.0, University of Verona, Italy
- 17/01/18** Julia: reinventing static analysis, University of Padova, Italy
- 17/01/18** Julia: an abstract interpretation-based static analyzer of industrial OO software, “Software Verification” course at University of Padova, Italy
- 06/11/17** Julia: an abstract interpretation-based static analyzer of industrial OO software, Max Planck Institute of Software Systems, Kaiserslautern, Germany
- 29/09/17** Scaling up Static Analyzers by Program Splitting, ETH, Zurich, Switzerland
- 10/11/16** Static Security Analysis for Mobile Software, ENISA Workshop “Mobile attacks and defense: from infrastructure to application”, Berlin, Germany
- 07/04/15** MorphDroid: Fine-grained Privacy Verification, Università Ca’ Foscari of Venice, Italy
- 31/03/15** A generic static analyzer and its application to TouchDevelop, Università della Svizzera Italiana, Lugano, Switzerland
- 22/04/14** Safety and cost analysis of TouchDevelop scripts, Università Ca’ Foscari of Venice, Italy
- 17/04/14** Hybrid Security Analysis of Web JavaScript Code via Dynamic Partial Evaluation, ETH Zurich, Switzerland
- 16/04/14** Static Security Analysis in an Industrial Setting: from Taint to Data-Centric Analysis, IBM Research Zurich, Switzerland
- 26/03/14** Structural Data Leakage, IBM Thomas J. Watson Research Center, Yorktown Heights, NY, USA
- 12/02/14** Secure Language Design and Program Analysis for Security (with Omer Tripp), Harvard University, Boston, MA, USA
- 15/10/12** Static analysis of multithreaded programs, IBM Thomas J. Watson research center, Yorktown, NY, USA
- 20/07/12** TVAL+: Combining TVLA and Value Analyses, Microsoft Research, Redmond, WA, USA
- 13/06/12** Inference of Fractional, Counting and Chalice Access Permissions via Abstract Interpretation, Universidad Complutense de Madrid, Madrid, Spain

12/06/12	TVLA and Value Analyses Together, Instituto Madrileño de Estudios Avanzados, Madrid, Spain
16/03/12	Automatic inference of access permissions, Department of Computer Science, Università Ca' Foscari, Venice, Italy
10/02/12	Access permission inference via abstract interpretation, Microsoft Research, Redmond, WA, USA
30/01/12	Automatic inference of fractional, counting and Chalice access permissions, IBM Watson research center, Hawthorne, NY, USA
25/02/11	Automatic inference of access permissions, Ecole Normale Supérieure of Paris, Paris, France
08/04/10	Checkmate: a Generic Static Analyzer of Java Multithreaded Programs, Department of Computer Science, Università Ca' Foscari, Venice, Italy
04/03/10	Abstract interpretation of memory models, IFIP WP 2.3, Lachen, Switzerland
25/08/09	Static analysis via abstract interpretation of multithread programs, IRISA-INRIA, Rennes, France
22/05/09	Static analysis via abstract interpretation of multithread programs, PhD Defense, Ecole Normale Supérieure, Paris, France
29/01/09	Static analysis by abstract interpretation of Java multithreaded programs, Chair of Programming Methodology, ETH, Zürich, Switzerland
18/12/08	Static analysis of the determinism of multithreaded programs, Lunch seminars, Università Ca' Foscari of Venice, Venice, Italy
20/06/08	Static analysis via abstract interpretation of the happens-before memory model, Ecole Normale Supérieure of Paris, Paris, France
28/05/08	Static analysis via abstract interpretation of the happens-before memory model, Lunch seminars, Università Ca' Foscari of Venice, Venice, Italy
19/12/07	Static analysis of unsafe code, Lunch seminars, Università Ca' Foscari of Venice, Venice, Italy
07/11/07	Static analysis of unsafe code, Microsoft Research, Redmond, WA, USA
16/02/06	JAIL (Javacard Abstract Interpretation-based lifeguard), Workshop AIDA, Pisa, Italy

OTHERS

11/05/22	Informatica: tra scienza e sviluppo del territorio, Open Day of the Master in Computer Science, Ca' Foscari University, Venice, Italy
08/04/22	Software Development and Engineering curriculum of the Master in Computer Science, Ca' Foscari University, Venice, Italy
02/10/20	Static Analysis for Discovering IoT Vulnerabilities, ISACA VI Venice AppSec, Venice, Italy
02/10/20	Static Analysis for Discovering IoT Vulnerabilities, ISACA VI Venice AppSec, Venice, Italy
28/03/19	"Start-up: imprenditoria giovane e innovativa", Associazione Scienze e Tecnica a Verona, Italy
13/03/19	"Hunting Software Security Vulnerabilities and Privacy Leaks with Semantic Static Analysis", Cloud & Cyber Security Expo, London, United Kingdom
05/03/19	"Semantic Static Analysis of IoT Software", VerifySoft Static Code Analysis Day, Offenburg, Germany
27/02/19	"Semantic Static Analysis of IoT Software", Embedded World 2019, Nuremberg, Germany
09/10/18	Hunting Software Security Vulnerabilities and Privacy Leaks with Semantic Static Analysis, the IT Security Expo and Congress (it-sa), Nuremberg, Germany
05/10/18	Analisi Statica per la Compliance GDPR, ISACA VI Venice AppSec, Venice, Italy

07/11/17	Code Analysis Reinvented: boosting Software Security and Privacy, W-JAX, Munich, Germany
17/10/17	Software security, reliability and privacy: what role can static analysis play?, 31st AIEA National Conference (ISACA Milan Chapter), Milan, Italy

Teaching

Student evaluation of courses at Ca' Foscari University

Course	19/20	20/21	21/22	22/23	23/24	24/25
Object-Oriented Programming		8.3	7.8	7.7	7.7	7.6
Software Architecture				7.7	8.5	8.4
Introduction to Coding and Data Management	7.2	7.6	5.9	6.5	7.0	6.0
Programming and Laboratory		7.9	7.7			

Spring sem. 2026	Lecturer , Typage et analyse statique. 35 hours of lectures.	Sorbonne Université, Paris, France
Spring sem. 2026	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2025	Lecturer , IT innovation for cultural organizations (part of the first level master in Management of Cultural Assets and Activities). 26 hours of lectures.	Ca' Foscari Challenge School, Venice, Italy
Fall sem. 2025	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2025	Lecturer , Software Architectures (part of the master in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Spring sem. 2025	Lecturer , Typage et analyse statique. 35 hours of lectures.	Sorbonne Université, Paris, France
Spring sem. 2025	Lecturer , Programmation comparée. 6 hours of lectures.	Sorbonne Université, Paris, France
Spring sem. 2025	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2024	Lecturer , IT innovation for cultural organizations (part of the first level master in Management of Cultural Assets and Activities). 26 hours of lectures.	Ca' Foscari Challenge School, Venice, Italy
Fall sem. 2024	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2024	Lecturer , Software Architectures (part of the master in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Spring sem. 2024	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2023	Lecturer , IT innovation for cultural organizations (part of the second level master in Management of Cultural Assets and Activities). 26 hours of lectures.	Ca' Foscari Challenge School, Venice, Italy
Fall sem. 2023	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2023	Lecturer , Software Architectures (part of the master in Computer Science). 48 hours of lectures.	Ca' Foscari University, Venice, Italy
Spring sem. 2023	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures. Evaluation: 7.13/10.	Ca' Foscari University, Venice, Italy
Fall sem. 2022	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures. Evaluation: 7.67/10	Ca' Foscari University, Venice, Italy
Fall sem. 2022	Lecturer , Software Architectures (part of the master in Computer Science). 48 hours of lectures. Evaluation: 7.65/10.	Ca' Foscari University, Venice, Italy
April 2022	Lecturer , A Guided Tour to Static Program Analysis: State-of-the-Art Tools and Techniques. 4 hours of lectures.	Global Initiative of Academic networks ⁶ , India
Spring sem. 2022	Lecturer , Programming and laboratory (part of the bachelor in Computer Science). 48 hours of lectures. Evaluation: 7.68/10.	Ca' Foscari University, Venice, Italy

⁶<https://gian.iitkgp.ac.in/>

Spring sem. 2022	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures. Evaluation: 5.57/10.	Ca' Foscari University, Venice, Italy
Fall sem. 2021	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures. Evaluation: 7.77/10	Ca' Foscari University, Venice, Italy
Spring sem. 2021	Lecturer , Programming and laboratory (part of the bachelor in Computer Science). 48 hours of lectures. Evaluation: 7.92/10 (course), 8.11/10 (professor).	Ca' Foscari University, Venice, Italy
Spring sem. 2021	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures. Evaluation: 7.63/10 (course), 7.62/10 (professor).	Ca' Foscari University, Venice, Italy
December 2020	Lecturer , Advances in Autonomous, Distributed and Pervasive Systems. 10 hours of lectures.	Ca' Foscari University, Venice, Italy
Fall sem. 2020	Lecturer , Object-Oriented Programming 1 (part of the bachelor in Computer Science). 48 hours of lectures. Evaluation: 8.26/10 (course), 8.38/10 (professor).	Ca' Foscari University, Venice, Italy
Spring sem. 2020	Lecturer , Introduction to Coding and Data Management 2 (part of the bachelor in Digital Management). 30 hours of lectures. Evaluation: 7.22/10 (course), 7.22/10 (professor)	Ca' Foscari University, Venice, Italy
Spring sem. 2020	Lecturer , Information System for the Arts (part of the master in Economics and Management of Arts and Cultural Activities). 30 hours of lectures. Evaluation: 7.21/10 (course), 8.13/10 (professor).	Ca' Foscari University, Venice, Italy
Spring sem. 2019	Lectures (4 hours) , Software Correctness, Security, and Reliability (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2018	Lectures (4 hours) , Software Correctness, Security, and Reliability (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2017	Lectures (4 hours) and evaluation of student exercises and final projects , Software Correctness, Security, and Reliability (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2012	Lecturer , Static Program Analysis. 13 lectures (2 hours each one with around 20 students) and oral exam preparation and evaluation	ETH Zürich, Switzerland
Spring sem. 2012	Lectures (4 hours) and evaluation of student projects , Analysis and Verification of Programs (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Fall sem. 2011	Teaching assistant , Concepts of Object-Oriented Programming (Prof. P. Müller). 13 exercise sessions (2 hours each one with around 20 students) and written exam preparation	ETH Zürich, Switzerland
Spring sem. 2011	Teaching assistant , Functional Programming and Formal Methods (Prof. P. Müller). Exercise sessions and written exam preparation	ETH Zürich, Switzerland
Fall sem. 2010	Head teaching assistant , Concepts of Object-Oriented Programming (Prof. P. Müller). 12 exercise sessions (2 hours each one with around 10 students) and written exam preparation	ETH Zürich, Switzerland
Fall sem. 2010	Lectures (8 hours) and evaluation of student projects , Analysis and Verification of Programs (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2009	Teaching assistant , Functional Programming and Formal Methods (Prof. P. Müller). Exercise sessions and written exam preparation	ETH Zürich, Switzerland
Fall sem. 2009	Teaching assistant , Concepts of Object-Oriented Programming (Prof. P. Müller). 4 exercise sessions (2 hours each one) with around 10 students. Exercise sessions and written exam preparation	ETH Zürich, Switzerland
Fall sem. 2009	Lectures (8 hours) and evaluation of student projects , Analysis and Verification of Programs (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2009	Lectures (8 hours) and evaluation of student projects , Analysis and Verification of Programs (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy
Spring sem. 2008	Lectures (16 hours) , Analysis and Verification of Programs (prof. A. Cortesi)	Ca' Foscari University, Venice, Italy

Student Supervision

Number of supervised theses per degree and year

Degree	2021	2022	2023	2024	2025	2026
BSc Computer Science		10	17	11	35	12
MSc Computer Science			2	1	6	5
BSc Digital Management	4	6	3	9	6	1
MSc other		2				

Academic year 2025/26	Supervisor, Master thesis of Almaz Gabdulbarov ("Development of a Web Player for Interactive Video", 100/110) Supervisor, Master thesis of Lorenzo Bastianello ("From legacy monoliths to [micro]services: design of an API gateway with centralized authentication", 100/110), and Filippo Vladimir Scapin ("Automated Deployment and Management of Scalable Kubernetes Clusters: A Zero-Touch Solution for On-Premises Virtualized Environments", 110/110 cum laude), Elia Bertapelle ("Orchestration Made Simple: A Practical Analysis of Aspire in a Cloud-Native Environment", 110/110), Marco Chinellato ("Automation of Banking Process: Document Processing System ", 110/110 cum laude), Sajede Salimi ("Static Analysis of REST Architecture in Spring Microservices with JLiSA", 82/110), and Soheil Izadiazarm ("Trace-Affinity for Scalable Observability Pipelines in Kubernetes Benchmarking ", 98/110)	<i>Master in Computer Science, University of Venice, Italy</i>
Academic year 2024/25		<i>Master in Computer Science, University of Venice, Italy</i>

Academic year 2024/25	Supervisor, Bachelor theses of Vincenzo Luca Biscotti ("Sviluppo e manutenzione di un'architettura Big Data per la gestione di flussi IoT in ambito GIS", 82/110), Sebastiano Boato ("Architettura di un'Interfaccia di Configurazione e Monitoraggio SCADA: Gestione del Flusso di Dati e Componenti Funzionali per l'Acquisizione Operativa IoT", 108/110), Alessandro Campagnaro ("Sistemi NLP per la classificazione di stringhe di diagnosi in ambito automotive", 98/110), Tudor Croitor ("User notifications with Kafka streams", 92/110), Tomislav Dzepina ("Database Blockchain for Medical Records Usage", 88/110), Claudio Giacomello ("Una soluzione per la gestione dei mezzi aziendali", 95/110), Elena Pozzebon ("Analisi e simulazione di attacchi a livello Data Link e Network: tecniche di difesa con ARP Guard", 108/110), Enrico Secchi ("Internalizzazione di Erglink", 91/110), Filippo Tiozzo ("Utilizzo di modelli NLP open source in locale: un approccio privacy-oriented con HuggingFace", 90/110), Giovanna Zamproga ("Dall'approccio euristico al Simulated Annealing: Ottimizzazione dell'inventario rotativo di magazzino", 106/110), Riccardo Comellato ("Dalle architetture monolitiche ai microservizi: Studio comparativo e implementazione pratica", 91/110), Federico Gaudenzi ("Chatbot: automatizzare l'assistenza aziendale", 95/110), Andrea Maggetto ("API development for a Real-Time Voice Cloning application", 90/110), Luca Menegazzo ("Migrazione di un Microservizio di Autenticazione in Rust", 107/110), Enrico Miatto ("SharePoint Online: Implementazione di una soluzione SaaS nella pubblica amministrazione", 104/110), Leonardo Montagner ("Come la programmazione ad oggetti permette lo sviluppo di applicazioni web in modo più robusto e strutturato rispetto alle metodologie contemporanee più utilizzate", 92/110), Michael Sarto ("IoT Ecosystem: From Physical Devices to Software and Network Infrastructure Management", 91/110), Alessia Tonello ("Sviluppo di test automatici con tecnologia REST", 84/110), Gabriele Torresan ("Sviluppo di un'applicazione mobile per la raccolta del latte utilizzando .NET MAUI", 108/110), Riccardo Vignotto ("I principi SOLID e i Design Patterns - Fondamenti teorici e applicazioni pratiche (con esempi in Python)", 98/110), Marco Bertocco ("Sviluppo di un software gestionale per industrie produttrici di tabacco", 92/110), Andrea Zanatta ("Reengineering an Industrial iOS Application into a Modern Web Platform Using Vue.js", 102/110), Lorenzo Gabor Mian ("Realizzazione di una procedura parallela e scalabile per il calcolo bollette dei rifiuti", 98/110), Diego Passarella ("Progettazione e sviluppo di un web tool di supporto decisionale per la valutazione del rischio dei materiali", 90/110), Andrea Musco ("Progettazione e realizzazione di una procedura di calcolo bollette per la tariffa dei rifiuti aventi le caratteristiche del parallelismo e della scalabilità", 100/110), Leonardo Zanatta ("Observability Monitoraggio delle applicazioni", 93/110), Alessandro Sozzo ("Design and Implementation of a Distributed Microservices Architecture for Agile KPI Intelligence", 94/110), Nichole Alagna ("Automated Load Simulation and Workload Monitoring in Agile Environments: A Web-Based Approach with Azure DevOps", 101/110), Pietro Donega ("ArcGIS per l'analisi territoriale e la produzione cartografica", 89/110), Matteo Pescarino ("Approcci Tradizionali e Piattaforme No-Code/Low-Code nello Sviluppo Web: Un Confronto Empirico", 89/110), Leonardo Gazzola ("Applicativo per la misura del formato di stampa di un film prodotto da una rotativa", 102/110), and Michael Basta ("A Scalable, Configuration-Driven Infrastructure for Distributed Web Scraping and High-Quality Dataset Construction", 86/110). Supervisor, Bachelor theses of Andrea Scordari ("From legacy monoliths to [micro]services: design of an API gateway with centralized authentication", 99/110), Cesare Junior Cims (94/110), Andrea Bernardo ("Il ruolo del design thinking nella progettazione di servizi digitali", 110/110 cum laude), Luca Camporese ("Process Mining as a Tool For Process Optimization", 106/110), Ester Girardi ("Artificial Intelligence in Manufacturing - A Case Study from the Printing Industry: Helix", 110/110 cum laude), and Ludovico Illotti ("Modernizing Enterprise Business Intelligence: A Case Study of Migration from a Legacy System to Microsoft Fabric", 88/110)	<i>Bachelor in Computer Science, University of Venice, Italy</i>
Academic year 2024/25	Supervisor, Bachelor theses of Andrea Scordari ("From legacy monoliths to [micro]services: design of an API gateway with centralized authentication", 99/110), Cesare Junior Cims (94/110), Andrea Bernardo ("Il ruolo del design thinking nella progettazione di servizi digitali", 110/110 cum laude), Luca Camporese ("Process Mining as a Tool For Process Optimization", 106/110), Ester Girardi ("Artificial Intelligence in Manufacturing - A Case Study from the Printing Industry: Helix", 110/110 cum laude), and Ludovico Illotti ("Modernizing Enterprise Business Intelligence: A Case Study of Migration from a Legacy System to Microsoft Fabric", 88/110)	<i>Bachelor in Digital Management, University of Venice, Italy</i>

Academic year 2023/24	Supervisor, Master thesis of Antonio Cimino ("CronFrame: A Macro Annotation Cron Job Framework with Web Server and CLI Tool written in Rust", 106/110), Marco, Biondo ("Ensuring data quality at the edge", 104/110), Mazhar Hussain ("Hackersgen: A Retrospective Analysis of Its Features, Architecture, and Development Practices", 92/110), Ubaid Ullah ("Development and Implementation of Urbana IoT Platform: Real-Time Analytics, Data Management, Processing, and Visualization for Scalable IoT Applications", 92/110), and Francesco Vinci ("Design and Implementation of a Report Management System for Antonio Zamperla S.p.A.", 101/110)	Master in Computer Science, University of Venice, Italy
Academic year 2023/24	Supervisor, Bachelor theses of Davide Cazzin ("Ingegneria del Software nelle Big Tech", 107/110), Andrea De Biasi ("Machine learning per l'analisi di batterie elettroniche", 104/110), Martina Ragusa ("HTTP: il Funzionamento di Richieste e Risposte con l'Utilizzo di Spring e Tomcat", 93/110), Mattia Schiavon ("Confronto tra Apache Kafka e Redpanda", 105/110), Lorenzo Silvello ("Deep Reinforcement Learning nell'industria 4.0", 100/110), Alberto Tomasin ("Optimizing Avian Influenza surveillance data reporting: a Public Health informatics tool for accurate data collection and transmission to EFSA", 93/110), Simone Dinato ("Innovations in Business Automation: From Robotic Process Automation to Hyperautomation", 97/110), Pietro Bonaventura ("Creazione e testing di un WebSocket con le tecnologie di Amazon Web Services", 98/110), Giacomo Femia Bertanno ("Applicazioni distribuite a microservizi", 99/110), Tommaso Soncin ("A Multi-Layered Approach to Network Security: Implementing WireGuard and Advanced Hardening Techniques", 101/110), Andrea Brion ("Automazione dei processi aziendali SuiteScript per la Semplificazione dei Flussi di Lavoro", 88/110), Enrico Brognera ("Sviluppo di workflow basati su reinforcement learning e digital twin per il supporto decisionale in ambito industriale", 103/110), Giovanni Coldebella ("Il Natural Language Processing tramite modelli generativi basati su Transformer", 85/110), Stefano Forin ("Development of an OPC UA Server for PLC structure simulation: an analysis of the design and implementation process for the pharmaceutical sector", 94/110), Rebecca Frisoni ("Analisi sulla Stabilizzazione e Ottimizzazione della Libreria SQLAlchemy di Zamperla: Tecniche di DB Diff e Accelerazione del Post-Processing", 90/110), Eleonora Tiziana Greco ("LLM e Fine-Tuning: Analisi e Ottimizzazione del Modello DeepSeek", 100/110), Marco Luisotto ("Creazione Plugin per Sottolineature di Sintassi", 86/110), Nicola Marizza ("Analisi di contenuti documentali tramite Optical Character Recognition e Large Language Models", 101/110), Luca Palmetti ("Strategie di Eco-Driving per Veicoli Autonomi: Reinforcement Learning per l'Ottimizzazione delle Prestazioni", 96/110), Marco Piazzon ("Simulatore di calcolo delle emissioni di anidride carbonica per il trasporto di materiale di ufficio", 93/110), Singh Baljinder ("Realizzazione di un portale Web per il conferimento di rifiuti", 88/110), Michele Trolese ("L'importanza di una password forte", 91/110), Filippo Zampiron ("Sviluppo Sito E-commerce", 87/110), Jinpeng Zhang ("Analisi del Ciclo di Vita di un Applicativo: un caso di studio nell'Ambito Pubblico", 84/110), and Luca Zuccarello ("WireGuardDashboard - Sviluppo di una dashboard per il monitoraggio di reti VPN", 89/110)	

Academic year 2023/24	Supervisor, Bachelor theses of Katherin Sofia Diaz Montiel ("AI Momentum in Campari: An IT Global Governance Perspective", 98/110), Christian Grossele ("Investigating Digital Transformation and Sustainable Practices in Fashion: OTB's Certilogo Project", 107/110), Laura Nicoletto ("Customer Relationship Management in the Context of Customer Experience: The Kering Eyewear Case", 110/110 cum laude), Marta Perissinotto ("Integrated Management of Intercompany Flows: Stiga S.p.A. case study", 98/110), Pizzo Mattia ("Sensitivity analysis and stress testing with forward-looking ML algorithms on credit risk", 95/110), Umberto Bolzoni ("Pricing Strategies in Italy's Large-Scale Retail Sector: The Role of Data and Agility ", 92/110), Gioele Squizzato ("Data Mesh Architecture: A Paradigm Shift in Retail Data Management", 98/110), David-Nicolae Burca ("Composable architecture in the hard machinery industry", 80/110), Kalist Filipovski ("Assessing the Impact of Customization on ERP Implementation Success: A Mixed-Methods Study of Microsoft Dynamics 365", 84/110), and Francesco De Bon ("Business Intelligence, from Data to Strategy Methodologies and tools for the management and interpretation of business data", 106/110)	<i>Bachelor in Digital Management, University of Venice, Italy</i>
2023-2026 (39th cycle)	Supervisor , PhD thesis of Federico Serafini (BUGSENG)	<i>Industrial PhD in Computer Science, University of Venice, Italy</i>
2023-2026 (39th cycle)	Supervisor , PhD thesis of Giacomo Boldini: "Static Analysis approaches for improving critical Software Engineering"	<i>PhD in Computer Science, University of Venice, Italy</i>
2023-2026 (39th cycle)	Supervisor , PhD thesis of Giacomo Zanatta: "Program Analysis for Robotics Software"	<i>PhD in Computer Science, University of Venice, Italy</i>
Academic year 2022/23	Supervisor, Master thesis of Giacomo Zanatta ("LiSA and rospy: Static Analysis for Robotics", 105/110), and Stefano Sello ("Zamperla Analytics Service: design and deployment of a microservice architecture with Kubernetes ", 100/110). Supervisor, Bachelor theses of Silvia Prata ("Gestione asincrona del sistema di download di dati da nodi edge e di upload in database tramite code di messaggi in Apache Kafka", 88/110), Giacomo Civiero ("Scaricare e archiviare la posta elettronica delle Scuole Pubbliche Italiane", 107/110), Riccardo Nalgi ("Analisi, identificazione dei punti di intervento e gestione degli errori di una applicazione e-commerce", 88/110), Giovanni Stevanato ("A web application for the game of chess", 97/110), Matteo Bordignon ("Integrazione MES aziendale progetto Industria 4.0", 100/110), Marco Chinellato ("Architettura software per il controllo di dati IIOT", 110/110 cum laude), Serena Zanon ("Verso un mondo serverless", 106/110), Enrico De Zorzi ("Integrazione di applicativi PHP in WordPress", 86/110), Riccardo Martignago ("Stream processing: caso d'uso in azienda", 100/110), Matteo Pagano ("Ottimizzazione dell'Architettura Software di Zamperla: Una transizione verso Microservizi e Sistema di Log Avanzato", 95/110), Marco Pistollato ("Progettazione e realizzazione di un sito web per la localizzazione di contenitori raccolta", 103/110), Alex Semenzato ("Implementazione di una strategia di monitoraggio della rete e applicazione della filosofia Zero Trust", 92/110), Tommaso Spolaor ("Gestione degli Errori di un'Applicazione Reale", 85/110), Marco Terrin ("Hardening of Zamperla's edge nodes", 101/110), and Matteo Emmerich ("Manutenzione del software Zamperla: Ottimizzazione attraverso Design Modulare e Integrazione di SQLAlchemy", 99/110). Supervisor, Bachelor theses of Silvia Casanova ("Digitalization inside the ERG Group: A Power Production Forecasts Combination Project", 106/110), Tommaso Pasut ("Data-Driven Auditing and Continuous Monitoring in the Banking Industry - the case of Illimity Bank", 107/110), and Subba Cecchetto ("Unraveling Communication Insights on Modern Society and the War in Ukraine", 94/110)	<i>Master in Computer Science, University of Venice, Italy</i>
Academic year 2022/23		<i>Bachelor in Computer Science, University of Venice, Italy</i>
07/2023 - 06/2024	Supervisor , Postdoc of Luca Negrini and Luca Olivieri on "Application of LiSA to robotic software"	<i>University of Venice, Italy</i>
From 05/2023	Mentor , PhD thesis of Sarah Tilscher on "Interactive Static Analysis"	<i>Technical University of Munich, Germany</i>
01/2023 - 12/2026	Co-Supervisor , PhD thesis of Chachar Badaruddin on the application of static analysis to blockchain technologies	<i>University of Venice, Italy</i>

Academic year 2021/22	Supervisor, Bachelor theses of Simone Biondo ("Analisi di Salesforce come PaaS e confronto con altre soluzioni ", 110/110 cum laude), Simone Bozzolan ("Sistema informatico per l'analisi, l'elaborazione e la visualizzazione di statistiche riguardanti partite di scacchi", 106/110), Andrea D'Attero ("Sicurezza e Performance delle Piattaforme Online", 99/110), Davide Finesso ("Access policies in SROS e SROS2 ", 103/110), Susanna Lorenzini ("LiSA Analysis: progettazione e sviluppo di un'applicazione web per l'esecuzione di analisi statiche con LiSA", 96/110), Jessica Daniele ("SpringLiSA: Progettazione e sviluppo di un'interfaccia web Spring per l'analizzatore statico LiSA", 96/110), Omar Tartaggia ("Realizzazione di una applicazione Android per l'interfacciamento ad una domotica casalinga", 87/110), Francesco Benetello ("Realizzazione di un Widget Android per un applicazione mobile di Domotica", 85/110), Davide Popovic ("Realizzazione di un'applicazione Android per la firma digitalizzata e l'incasso dei documenti, la gestione di vuoti a rendere e la gestione dei resi in merce", 93/110), and Umberto Soffia ("Acquisizione dati in un ecosistema industriale", 99/110).	Bachelor in Computer Science, University of Venice, Italy
Academic year 2021/22	Supervisor, Bachelor theses of Marco Battiston ("Report Automation for industrial performance KPIs monitoring: an application in the Airport Sector ", 108/110), Riccardo Ceolin ("How to move an elephant", 106/110), Annalisa Laini ("Tackling Welfare: Digital Solutions in the Canton of Ticino ", 104/110), Leonardo Provenzano ("Path to conversions: How attribution models improve digital marketing channels performances", 109/110), and Galfione Alessandro ("Data Quality Analysis on Real Estate Data", 96/110).	Bachelor in Digital Management, University of Venice, Italy
Academic year 2021/22	Supervisor, Master thesis of Laura Coppelli (Crypto-art and NFTs: a technical and aesthetic analysis, 110/110 cum laude).	Master in Economics and Management of Arts and Cultural Activities, University of Venice, Italy
Academic year 2020/21	Supervisor, Bachelor theses of Mattia Vighesso (Monitoraggio applicativo distribuito: la soluzione di Previnet S.p.A., 107/110), Alessio Campanelli ("Visualizzazione dei risultati di analisi di LiSA", 110/110 cum laude) and Mario Gottardo ("Una dashboard per l'assessment e mitigazione del rischio in contesti DevOps", 109/110).	Bachelor in Computer Science, University of Venice, Italy
04/2022 - 10/2023	Supervisor , Postdoc of Ruffin White on "Application of static analysis to data processing notebooks"	University of Venice, Italy
10/2022 - 12/2022	Supervisor , Postdoc of Maikel Lazaro Perez Gort	University of Venice, Italy
Academic year 2020/21	Supervisor , Master thesis of Elisa Paladin (Managing Museums Collections: the Role of the Registrar and the Use of Collections Management Systems (CMSs), 108/110).	Master in Economics and Management of Arts and Cultural Activities, University of Venice, Italy
12/2021 - 12/2022	Supervisor , Postdoc of Gianluca Caiazza on "Access policy inference and verification through static analysis"	University of Venice, Italy
Academic year 2020/21	Supervisor , Bachelor theses of Rebecca Galassi (RFM analysis and data-driven personas: how a company can improve the knowledge of the customer base for maximization of marketing performance, 107/110), Leonardo Levisse (The key role digital platforms have in influencing the positioning of a b2b firm, 101/110), Giulio Piccolo (Customer segmentation analysis of an ICT multinational company: a machine learning approach, 109/110), and Pietro Querci (Customers segmentation in the financial sector: a data drive approach, 110/110 cum laude).	Bachelor in Digital Management, University of Venice, Italy
Academic year 2019/20	Co-supervisor , Master thesis of Sara Ferro on software verification of PLC programs. Supervisor: prof. Agostino Cortesi	University of Venice, Italy
11/2017 - 02/2018	Co-supervisor , Master thesis of Luca Olivieri on frontend for GDPR analyses. Supervisor: prof. Fausto Spoto	University of Verona, Italy
10/2017 - 02/2018	Co-supervisor , Master thesis of Luca Negrini on advanced algorithms and application of machine learning to application splitting. Supervisor: prof. Fausto Spoto	University of Verona, Italy
03/2017 - 06/2017	Supervisor , Internship of Dinu Berinde on the application of Julia's analyses to Android programs	University of Verona, Italy
10/2012 - 03/2013	Supervisor , Master thesis at ETH of Zürich of Daniel Schweizer ("Overapproximating the cost of loops in TouchDevelop")	ETH Zürich, Switzerland
03/2011 - 08/2011	Supervisor , Bachelor thesis at ETH of Zürich of Raphael Fuchs ("Interfacing TVLA and Sample")	ETH Zürich, Switzerland

03/2011 - 08/2011	Supervisor , Master thesis at ETH of Zürich of Dominik Gabi (“Disjunction on demand”)	ETH Zürich, Switzerland
01/2010 - 06/2010	Supervisor , Bachelor thesis of Roman Scheidegger (“Translating Java bytecode to Simple”)	ETH Zürich, Switzerland
12/2019 - 05/2010	Co-supervisor , Master thesis of Giulia Costantini (“Abstract domains for static analysis of strings”, mark: 110/110 cum laude). Supervisor: prof. Agostino Cortesi	Ca’ Foscari University, Venice, Italy

References

Patrick Cousot (pcousot@cs.nyu.edu), Full professor at Courant Institute of Mathematical Sciences	New York University, U.S.A
Peter Müller (peter.mueller@inf.ethz.ch), Full professor and head of the Chair of Programming Methodology	ETH, Zürich, Switzerland
Agostino Cortesi (cortesi@unive.it), Full professor at the Department of Environmental Sciences, Informatics and Statistics	Ca’ Foscari University, Venice, Italy
Fausto Spoto (fausto.spoto@univr.it), Associate professor at the Department of Computer Science	University of Verona, Italy

Other Titles

21/06/05	Admitted in the Italian Register of engineers
21/01/05	Obtained the European Computer Driving License
20/02/03	Admitted in the Register of journalists as occasional contributor

Hobbies

Chess	I am a decent chess player ⁷) and chess instructor, vice-president of the Battinelli chess club in Verona, member of its governing council, and captain of one of its teams. I have been playing chess since I was a child, and I have been playing for Capablanca club in Mestre (Venice), Paris Université Club in Paris, and Reti Club in Zurich (where I was the captain of one of the teams).
Journalism	In relatively old times (that is, about 15 years ago) I have been an active sport journalist for several newspapers, websites, radios and televisions. I am part of the Register of journalists as occasional contributor since March 2003.
Mountains	I love skiing (https://mydolomiti.dolomitisuperski.com/profilopubblico.aspx?p=45A593FB816B5E03E0537D1DC7D95559 contains my ski performances of the last years - accessible only by logged users) and hiking (https://connect.garmin.com/modern/activity/3844672704 is a recent hike (summer 2019) I am particularly proud of), especially in the Dolomiti mountains.

⁷<http://ratings.fide.com/card.phtml?event=673129>